Reducing Floats mod p

Algebraic FPSan as a residue semantics

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Abstract

FPSan is a dynamic testing technique for numerical programs: replace each floating-point value by a same-width fingerprint, run the program in a finite target ring, and compare the resulting fingerprints. The original Triton FPSan introduced this idea using a tuned bijection from source-format bit patterns into $\mathbb{Z}/2^w\mathbb{Z}$, where w is the source width [1, 2]. That scramble is useful, but it is not a homomorphism from the represented numeric values, so identities such as $2 + 2 = 4$ or $x + x = 2x$ need not hold.

This paper studies algebraic FPSan: a homomorphic alternative to Triton's FPSan scramble on finite binary floating-point values. The observation is simple: every finite binary float is a dyadic rational, hence an element of $\mathbb{Z}[1/2]$. For every odd prime p, reduction modulo p gives a ring homomorphism

$$\mathbb{Z}[1/2] \longrightarrow \mathbb{F}_p.$$

This gives a finite-field residue semantics for FPSan that preserves exact rational identities while keeping the same-width fingerprint invariant. We then study the constraints on the prime p, multiplicative casts between widths, composite residue rings for selected exponential, logarithmic, and trigonometric laws, multi-prime recovery by the Chinese remainder theorem (CRT), and extensions to algebraic-number formats.

1 Introduction

Suppose a numerical kernel is rewritten for performance. A reduction may be reassociated, an expression may be factored, an FMA may replace a multiply and an add, or a matrix intrinsic may change the accumulation order. Comparing floating-point results bit-for-bit is too strict: a value-preserving algebraic rewrite can change rounding. Tolerance tests, on the other hand, are often too weak and too problem-specific.

Triton's FPSan introduced a useful third option [1]. Instead of asking whether two rounded floating-point executions are bitwise equal, FPSan asks whether the two computations have the same fingerprint in a chosen target ring. Each source value is replaced by a fingerprint, and each source operation is replaced by a deterministic operation on fingerprints. The resulting semantics are designed to be compact and invariant under formal laws such as associativity and distributivity. They are not a reference implementation of floating-point rounding.

The invariant throughout this document is width preservation. If the original floating-point type has w bits, then every FPSan semantics discussed here stores its fingerprint in exactly w bits. The central design choice is the target ring together with the leaf map. One chooses a target ring encodable in the same bit width, possibly with distinguished sentinel codes adjoined to the finite ring, and one chooses how each source floating-point value is sent into that target ring.

Triton's concrete semantics instantiate this design by choosing the target ring $\mathbb{Z}/2^w\mathbb{Z}$ and a tuned bijection, not a ring homomorphism,

$$E : \{\text{source-format bit patterns of width } w\} \longrightarrow \mathbb{Z}/2^w\mathbb{Z}.$$

The construction itself is mostly indifferent to the details of a particular floating-point standard: what matters is that the sanitizer receives fixed-width bit patterns. Writing 0 and 1 for the bit patterns of those values, and writing $-x$ for source-format sign negation of a bit pattern x, Triton's tuning gives $E(0) = 0$, $E(1) = 1$, and $E(-x) = -E(x)$ in the target ring. After applying E, arithmetic is performed with ring operations in $\mathbb{Z}/2^w\mathbb{Z}$, with special handlers for operations such as division, exponentials, and trigonometry. The Triton author's blog discusses the algebraic intentions of this construction [2].

Because E is a bijection, one can always define an artificial operation on bit patterns by transporting the target-ring operation back through E:

$$x \boxplus_E y = E^{-1}(E(x) + E(y)).$$

With respect to $\boxplus_E$, the map E is a homomorphism by construction. That statement is true but not the statement a compiler engineer usually wants: $\boxplus_E$ is not floating-point addition, nor exact rational addition of the represented values. It is the arithmetic manufactured by the embedding.

The visible consequence is immediate. If u is the source-format bit pattern of 2.0, then typically

$$E(u) + E(u) \neq E(\text{bit pattern of } 4.0).$$

So Triton FPSan sees reassociation and distributivity, but it does not see ordinary value facts such as $2 + 2 = 4$, $x + x = 2x$, or $x/x = 1$ for every nonzero x.

The contribution of algebraic FPSan is that the opposite leaf-map choice is possible. We can keep compact same-width fingerprints, but choose the leaf map to be a ring homomorphism on exact finite values. The price is that a single same-width reduction is no longer injective. The benefit is that its collisions are ordinary modular collisions in a characteristic-zero calculation reduced modulo a prime, not artifacts of an unrelated transported arithmetic. The next section gives the algebraic observation that makes this possible.

2 Finite binary floats live in $\mathbb{Z}[1/2]$

A finite binary floating-point value can be written as

$$x = 2^e M,$$

where $M \in \mathbb{Z}$ is a signed integer significand and $e \in \mathbb{Z}$. Normal and subnormal values both have this form, after absorbing the sign into M. Thus every finite value lies in

$$\mathbb{Z}[1/2] = \left\{ \frac{a}{2^k} : a \in \mathbb{Z},\ k \geq 0 \right\}.$$

This is the localization of $\mathbb{Z}$ obtained by inverting 2.

The maximal ideals of $\mathbb{Z}[1/2]$ are exactly the ideals generated by odd primes. Equivalently, an odd prime p is precisely a prime modulo which 2 remains invertible. The resulting residue homomorphism

$$\mathbb{Z}[1/2] \longrightarrow \mathbb{F}_p$$

is the leaf map used by the Field-family semantics: a finite source value is sent to its residue in the target field. Concretely, this homomorphism is determined by the image of $1/2$, which is taken to be the inverse of 2 in $\mathbb{F}_p$. This is the only algebraic fact about finite binary floats that the Field-family semantics need.

This is the mathematical core of the Field-family semantics. The fingerprint of a finite value is the residue of its exact dyadic value. Addition, subtraction, and multiplication are the field operations. The faithful Field variants use field inversion for division; `FieldFast` keeps the same finite-value fingerprints and the same additive/multiplicative arithmetic, but treats nontrivial division and roots as tagged operations for speed. The cheap division cases $0/x = 0$ and $x/1 = x$ are still preserved.

3 Compactness, sentinels, and non-injectivity

By width preservation, a source format of width w gives only 2^w fingerprint codes. Let N be the finite residue modulus used by an algebraic semantics; for example, in the `Semantics::Field` case with target $\mathbb{F}_p$, $N = p$, while in the composite ring cases $N = pd$. The finite residues use the codes $0, \ldots, N-1$, and two further codes are reserved for non-finite values:

$$N \quad \text{for infinity,} \qquad N+1 \quad \text{for NaN.}$$

Thus the finite modulus must satisfy $N \le 2^w - 2$.

This compactness has a price: the algebraic map is not injective. It cannot be, because many finite floats must share a finite set of residues. This is unlike Triton's bit-pattern scramble, which is a bijection before arithmetic starts.

The following fact is what makes that compromise much less dangerous at the most important point.

Proposition 1 (Zero is not a finite collision)**.** *Let N be odd, and suppose every finite value of the source format can be written as $2^e M$ with $|M| < N$. Then its residue modulo N is zero if and only if the represented value is zero.*

Proof. Since N is odd, 2 is a unit modulo N. Hence

$$2^e M \equiv 0 \pmod{N} \quad \Longleftrightarrow \quad M \equiv 0 \pmod{N}.$$

The hypothesis $|M| < N$ makes the latter congruence equivalent to $M = 0$. That is exactly the zero value; the two IEEE-754 signed zeros are deliberately identified by the algebraic semantics. □

Nonzero collisions remain ordinary finite-modulus fingerprint collisions. If two distinct exact dyadic expressions on the same inputs compare equal, then after clearing powers of 2, their difference has numerator divisible by the chosen modulus. The suffix-2 variants exist so that a suspected collision can be checked against a different set of primes or composite moduli: `Field2`, `FieldFast2`, `SophieGermainRing2`, and `PythagoreanRing2` have entirely different nonzero collision sets.

4 Several primes and characteristic-zero recovery

The non-injectivity of a single reduction should not be confused with loss of the characteristic-zero arithmetic. If $p_1, \ldots, p_r$ are distinct odd primes and $P = \prod_i p_i$, then the combined map

$$\mathbb{Z}[1/2] \longrightarrow \prod_i \mathbb{F}_{p_i}$$

is, by the CRT, the same as reduction modulo P, with 2 still invertible. Thus several algebraic FPSan runs can be interpreted as a single wider residue computation. If an exact dyadic result is known to be $a/2^k$ with $|a| < P/2$, then the residues determine a uniquely by the symmetric CRT representative, and hence determine the dyadic value itself.

This is a fundamental distinction from Triton's FPSan. Triton's leaf map is invertible, but after that the arithmetic is the arithmetic of the scrambled ring $\mathbb{Z}/2^w\mathbb{Z}$. There is no compatible family of reductions whose CRT lift is the original characteristic-zero calculation. Algebraic FPSan goes the other way: one prime is not injective, but different primes are all shadows of the same calculation in $\mathbb{Z}[1/2]$.

As a concrete bound, consider OCP FP8 E5M2 [3]. It has exponent bias 15 and two mantissa bits. Every finite value is $A/2^{16}$, and the maximum finite magnitude is

$$1.75 \cdot 2^{15} = 7 \cdot 2^{13},$$

so

$$|A| \leq 7 \cdot 2^{29} = 3\,758\,096\,384.$$

To reconstruct an arbitrary finite FP8 E5M2 value by signed CRT it is enough to have $P > 2 \cdot 7 \cdot 2^{29} = 7\,516\,192\,768$. If each residue is still required to fit in an 8-bit fingerprint with two sentinels, then $p \leq 254$. The four largest possible primes give

$$251 \cdot 241 \cdot 239 \cdot 233 = 3\,368\,562\,317,$$

which is not enough; no four-prime set can do better. Five primes suffice, for example

$$251 \cdot 241 \cdot 239 \cdot 233 \cdot 229 = 771\,400\,770\,593.$$

Thus five 8-bit prime reductions are cardinality-minimal for reconstructing any finite rounded FP8 E5M2 value. If one also insists on the Field-family root congruence $p \equiv 11 \pmod{12}$, five primes still suffice, for example

$$251,\ 239,\ 227,\ 191,\ 179.$$

The present library does not implement multi-prime reconstruction; its goal is compact same-width fingerprinting. The point is conceptual: algebraic FPSan collisions are ordinary modular collisions, and ordinary CRT arithmetic can, in principle, remove them within any known magnitude bound. Rounding is a separate semantic choice: the algebraic fingerprints track the exact dyadic expression between whatever rounded values the instrumented program presents as leaves.

5 Prime choices for the Field-family semantics

The statement "reduce modulo an odd prime" leaves many possible primes. The implementation chooses primes satisfying several constraints simultaneously. These constraints are motivated by different public semantics: some are needed by the faithful `Field` root operations, some by `FieldWithMulCasts` casts, and some by all Field-family modes. The prime tables are nevertheless shared by `Field`, `FieldFast`, and `FieldWithMulCasts` (and by their suffix-2 variants) so users can switch among these policies without changing the finite fingerprints of inputs.

5.1 Size and sentinel constraints

The prime should be close to 2^w, because a larger field gives a lower collision probability. It must still leave room for infinity and NaN:

$$p \leq 2^w - 2.$$

It must also be larger than the maximum possible $|M|$ for the format's signed significand M, so that Proposition 1 applies and zero is hit only by true zero.

Constraint 1 (size, sentinels, and zero)**.** *For a source format of width w, the Field prime p must satisfy*

$$M_{\max}(w) < p \leq 2^w - 2,$$

where $M_{\max}(w)$ is the largest possible $|M|$ in the representation $x = 2^e M$ for that format.

5.2 Algebraic roots

The faithful algebraic modes try to treat square root and cube root as algebraic power maps when the chosen modulus allows it, but the exact contract differs by semantics. In the Field family, Constraints 2 and 3 are used by `Field` and `FieldWithMulCasts`. `FieldFast` shares the same primes for compatibility, but deliberately represents roots by tagged fingerprints.

For `Semantics::Field`, the modulus is a prime p, so the unit group has order $p - 1$. The square-root power map uses the following condition.

Constraint 2 (square-root power map)**.**

$$p \equiv 3 \pmod 4.$$

This makes 2 invertible in the odd part of the unit-group exponent and gives the multiplicative square-root choice used below.

Under this condition,

$$S(x) = x^{(p+1)/4}$$

is a multiplicative square-root choice:

$$S(xy) = S(x)S(y).$$

It is a true inverse to squaring on the square residues and gives a consistent choice on the nonsquares. No multiplicative square root can be a two-sided inverse on all of $\mathbb{F}_p^\times$, since squaring is two-to-one.

For cube roots, the power map $x \mapsto x^r$ is a perfect cube root exactly when $3r \equiv 1 \pmod{p-1}$. That requires $3 \nmid p - 1$, i.e. $p \equiv 2 \pmod 3$.

Constraint 3 (perfect cube-root power map)**.**

$$p \equiv 2 \pmod 3.$$

Equivalently, 3 is invertible modulo $p - 1$, so a power map can be an inverse to cubing on $\mathbb{F}_p^\times$.

Together, Constraints 2 and 3 give

$$p \equiv 11 \pmod{12}.$$

That congruence is the root-related constraint on the Field primes.

The composite algebraic modes use the same power-map idea with the exponent of the unit group of $\mathbb{Z}/N\mathbb{Z}$. The Sophie Germain choices keep 3 invertible in that exponent, so `cbrt` remains a perfect power-map cube root. The Pythagorean choices cannot keep that property while also making trigonometric rotations available; there `cbrt` is represented as a tag. Square root remains a multiplicative power map in the faithful algebraic modes, although the round trip $\sqrt{x^2} = x$ holds only on the part of the ring selected by that power-map choice.

5.3 Multiplicative casts

The optional multiplicative-cast policy is the only cast policy here that tries to preserve algebraic structure. It is exposed as

`Semantics::FieldWithMulCasts`.

The ordinary `Semantics::Field` variants and `FieldFast` use the same primes and finite-value fingerprints, but cheap deterministic casts. Triton's casts are signed integer resizes of the scrambled fingerprint, and the composite algebraic semantics use a deterministic finite-residue convention.

A cast between different Field widths cannot be additive, hence cannot be a ring homomorphism. Indeed, if $p \neq q$, every additive-group homomorphism $(\mathbb{F}_p, +) \to (\mathbb{F}_q, +)$ is trivial. The only nontrivial algebraic structure available across different characteristics is therefore multiplicative: the groups of nonzero residues.

The multiplicative construction below applies only to finite nonzero fingerprints. Zero is mapped to zero, and the infinity/NaN sentinels map to the corresponding sentinels. The `FP32↔FP64` edge uses Pohlig–Hellman discrete logs because a linear scan over a group of size about 2^{32} would be too expensive; the group orders are chosen to make this feasible.

Constraint 4 (computable cast logarithms)**.** *The Field group orders used for cast logarithms, especially $m_{32} = p_{32} - 1$, must be smooth enough for the Pohlig–Hellman path used by the implementation.*

The multiplicative group $\mathbb{F}_p^\times$ is cyclic. The multiplicative-cast tower consists of the widths $4, 8, 16, 32, 64$; the 6-bit format uses a standalone prime and is not part of this tower. Write p_i for the Field prime at a tower width and $m_i = p_i - 1$. For each format class α at that width, choose a primitive root $g_{i,\alpha}$ of $\mathbb{F}_{p_i}^\times$. There is usually only one such class; the important exceptions are the two FP8 families, E4M3 and E5M2, and the two 16-bit families, FP16 and BF16. The implementation uses g_i for one class and g_i^3 for the other. Since the Field primes are $11 \pmod{12}$, 3 is coprime to m_i, so g_i^3 is again a primitive root. Let

$$L_{i,\alpha} : \mathbb{F}_{p_i}^\times \to \mathbb{Z}/m_i\mathbb{Z}$$

be discrete logarithm to the base $g_{i,\alpha}$. In these coordinates, multiplication in $\mathbb{F}_{p_i}^\times$ becomes addition in $\mathbb{Z}/m_i\mathbb{Z}$.

The chosen primes satisfy

$$m_4 \mid m_8 \mid m_{16} \mid m_{32} \mid m_{64}.$$

Moreover, the factors introduced at successive steps are coprime to the factors already present. Equivalently, whenever $m_i \mid m_j$, the cofactor m_j/m_i is coprime to m_i, and stepwise cofactors between i and j are pairwise coprime.

Constraint 5 (multiplicative cast tower)**.** *The Field primes in the cast tower must satisfy*

$$m_4 \mid m_8 \mid m_{16} \mid m_{32} \mid m_{64}, \qquad m_i = p_i - 1,$$

and whenever $i < j$, the cofactor m_j/m_i must be coprime to m_i; in particular, the stepwise cofactors from i to j are pairwise coprime.

Here and below, $i < j$ means that the i-bit tower format is narrower than the j-bit tower format. Put $c_{ij} = m_j/m_i$. The coprimality condition in Constraint 5 gives a Chinese-remainder section

$$\sigma_{ij} : \mathbb{Z}/m_i\mathbb{Z} \to \mathbb{Z}/m_j\mathbb{Z}$$

characterized by

$$\sigma_{ij}(a) \equiv a \pmod{m_i}, \qquad \sigma_{ij}(a) \equiv 0 \pmod{c_{ij}}.$$

The widening ("up") cast $U_{i\alpha,j\beta}$ and narrowing ("down") cast $D_{j\beta,i\alpha}$ are the multiplicative maps defined in logarithm coordinates by

$$L_{j,\beta}(U_{i\alpha,j\beta}(x)) = \sigma_{ij}(L_{i,\alpha}(x)), \qquad L_{i,\alpha}(D_{j\beta,i\alpha}(y)) = L_{j,\beta}(y) \bmod m_i.$$

For two distinct format classes α, β at the same width, the same logarithm-coordinate rule gives the automorphism

$$L_{i,\beta}(C_{i\alpha,i\beta}(x)) = L_{i,\alpha}(x).$$

Thus FP16 and BF16, or E4M3 and E5M2, are not silently identified even though they share a bit width and a Field prime.

Proposition 2 (The Field cast tower)**.** *For $i < j < k$, and for any choices of format classes at those widths, Field widening maps compose, Field narrowing maps compose, and narrowing after widening is the identity. Suppressing the format labels in the notation,*

$$U_{jk}U_{ij} = U_{ik}, \qquad D_{ji}D_{kj} = D_{ki}, \qquad D_{ji}U_{ij} = \mathrm{id}.$$

Also $D_{kj}U_{ik} = U_{ij}$. In the opposite order, $U_{ij}D_{ji}$ is generally not the identity on $\mathbb{F}_{p_j}^{\times}$; it is the projection onto the chosen order-m_i subgroup.

Proof. The maps U_{ij} and D_{ji} are homomorphisms because they are additive in logarithm coordinates. The narrowing identities are just transitivity of reduction modulo $m_i \mid m_j \mid m_k$. The widening identity follows from uniqueness in the CRT description: stepwise widening from i to j to k gives a logarithm congruent to the original one modulo m_i, and congruent to zero modulo each step cofactor; because those cofactors are pairwise coprime, this is the same as being zero modulo the total cofactor m_k/m_i, which is the direct widening U_{ik}. Similarly, $D_{kj}U_{ik}$ has logarithm $\sigma_{ik}(a) \bmod m_j$, which is congruent to a modulo m_i and to zero modulo m_j/m_i; by uniqueness this is $\sigma_{ij}(a)$, hence $D_{kj}U_{ik} = U_{ij}$.

The identity $D_{ji}U_{ij} = \mathrm{id}$ follows because $\sigma_{ij}(a) \equiv a \pmod{m_i}$. Finally, $U_{ij}D_{ji}$ first forgets the logarithm modulo the cofactor c_{ij} and then chooses the CRT lift with cofactor coordinate zero, so it is a projection, not the identity in general. The coprime-cofactor hypothesis is exactly what is used here: it gives the CRT sections σ_{ij}, and it turns "zero modulo each step cofactor" into "zero modulo their product," so stepwise widening agrees with direct widening. □

5.4 Actual constants

The actual Field-family prime choices are as follows. Variant 1 is used by `Field`, `FieldFast`, and `FieldWithMulCasts`; variant 2 is used by `Field2`, `FieldFast2`, and `FieldWithMulCasts2`. The 4-bit

prime is shared because it is the only 11 mod 12 prime at that width. The 6-bit prime is standalone and is not part of the cast tower.

	variant 1		variant 2		cast tower
w	p	g	p	g	
4	11	2	11	2	yes
6	59	2	47	5	standalone
8	191	19	131	2	yes
16	65171	2	64871	7	yes
32	4284862331	2	4291215371	2	yes
64	18446743887391934171	2	18446743217995397111	7	yes

Here g is the primitive root used for multiplicative casts in `FieldWithMulCasts`. At a bit width with two supported format classes, the implementation uses g for the first class and g^3 for the second: FP8 E4M3 uses g, FP8 E5M2 uses g^3, FP16 uses g, and BF16 uses g^3. Since every Field-family prime is 11 mod 12, 3 is coprime to $p-1$, so g^3 is again a primitive root. This is the concrete mechanism that makes same-width casts such as FP16↔BF16 visible to `FieldWithMulCasts` instead of collapsing them to the identity.

6 Why finite fields cannot model exponential laws

In any finite field $\mathbb{F}_q$, with q any prime power, any group homomorphism

$$(\mathbb{F}_q, +) \longrightarrow (\mathbb{F}_q^\times, \cdot)$$

is trivial: the source has order q, while the target has order $q-1$. Their orders are coprime. Thus a field-valued fingerprint cannot have a nontrivial function satisfying

$$\exp(a+b) = \exp(a)\exp(b)$$

as a function only of its field residue.

To model the exponential law, the fingerprint must remember another residue modulo a prime d for which the multiplicative side has an order-d subgroup, alongside a prime p whose unit group contains that subgroup. The Chinese Remainder Theorem keeps this compact. With $n = pd$,

$$\mathbb{Z}/n\mathbb{Z} \simeq \mathbb{F}_p \times \mathbb{F}_d.$$

The cost is algebraic: $\mathbb{Z}/n\mathbb{Z}$ has zero-divisors, so it is a ring rather than a field.

7 The Sophie Germain ring

`Semantics::SophieGermainRing` takes its target ring to be $\mathbb{Z}/n\mathbb{Z}$. The modulus has the form

$$n = pd, \qquad p = 2d + 1,$$

with p and d prime. Strictly speaking, d is the Sophie Germain prime and p is the corresponding safe prime; the semantics name refers to the pair.

For widths below 8, the implementation does not use a composite Sophie Germain ring. There is too little same-width room for two sentinels and a useful CRT exp/log channel. Instead, the 4- and 6-bit cases use the corresponding Field-family prime: the target is just $\mathbb{F}_p$, the auxiliary d-channel is

absent, and exp/log operations are represented as tags. Thus sub-8-bit `SophieGermainRing` behaves like the faithful Field semantics for finite algebraic arithmetic, but not for exp/log laws.

At widths $8, 16, 32, 64$, the actual Sophie Germain choices are as follows; variant 1 is the unsuffixed semantics and variant 2 is `SophieGermainRing2`.

variant	w	d	$p = 2d + 1$	$n = pd$
1	8	11	23	253
2	8	5	11	55
1	16	179	359	64261
2	16	173	347	60031
1	32	46229	92459	4274287111
2	32	46199	92399	4268741401
1	64	3037000121	6074000243	18446739472945029403
2	64	3036999209	6073998419	18446728393970250571

Since $d \mid p - 1$, the group $\mathbb{F}_p^\times$ contains a subgroup of order d. The implementation chooses an element $g \in (\mathbb{Z}/n\mathbb{Z})^\times$ whose $\mathbb{F}_p$-component has order d and whose $\mathbb{F}_d$-component is 1. It then defines

$$\exp(x) = g^{x \bmod d}.$$

This is a genuine homomorphism from the additive d-channel to the multiplicative order-d subgroup:

$$\exp(a + b) = \exp(a) \exp(b).$$

The logarithm is the dual map. It projects a unit to the order-d subgroup in the $\mathbb{F}_p$-component, takes the discrete logarithm to base g, and returns the corresponding element of the additive order-d subgroup of $\mathbb{Z}/n\mathbb{Z}$. Thus, on the modeled image,

$$\log(xy) = \log(x) + \log(y), \qquad \exp(\log(\exp x)) = \exp x.$$

The same channel carries other bases:

$$\exp_b(x) = g^{K_b x \bmod d}.$$

The constants K_b are fixed units modulo d. They are not numerical approximations to $\log_b e$; those constants are irrational and have no meaning as residues of dyadic values. The contract is that each base preserves its own law and has its own inverse.

Because the modulus is composite, division is exact only on units. If the denominator is a zero-divisor, the implementation returns the algebraic NaN fingerprint. At the sizes used here, zero-divisors are rare compared with the zero-divisors in Triton's $\mathbb{Z}/2^w\mathbb{Z}$, but they are real.

8 The Pythagorean ring

`Semantics::PythagoreanRing` takes its target ring to be $\mathbb{Z}/n\mathbb{Z}$. The modulus has the form

$$n = pd, \qquad p = 4d + 1,$$

again with p and d prime. This still gives $d \mid p - 1$, so the exp/log channel remains available. The new feature is that $p \equiv 1 \pmod 4$, so -1 has a square root in $\mathbb{F}_p$. This allows a finite-ring analogue of complex rotations in the p-component of the CRT decomposition.

For widths below 8, the implementation does not use a composite Pythagorean ring. As in the Sophie Germain case, the 4- and 6-bit cases use the corresponding Field-family prime, with

no auxiliary d-channel. Exp/log and sin/cos operations are therefore represented as tags at those widths. The usual Pythagorean tradeoff discussed below, including the loss of perfect `cbrt`, begins only at width 8; below that, finite algebraic arithmetic and roots follow the Field-family behavior.

At widths $8, 16, 32, 64$, the actual Pythagorean choices are as follows; variant 1 is the unsuffixed semantics and variant 2 is `PythagoreanRing2`.

variant	w	d	$p = 4d + 1$	$n = pd$
1	8	7	29	203
2	8	3	13	39
1	16	127	509	64643
2	16	97	389	37733
1	32	32707	130829	4279024103
2	32	32647	130589	4263339083
1	64	2147483059	8589932237	18446733956915472983
2	64	2147480707	8589922829	18446693549896360103

The implementation writes the construction in the quadratic algebra $(\mathbb{Z}/n\mathbb{Z})[i]$ with a formal $i^2 = -1$. Here $\mathbb{F}_p$ should be read as a CRT factor of $\mathbb{Z}/n\mathbb{Z}$, not as a unital subring:

$$\mathbb{Z}/n\mathbb{Z} \simeq \mathbb{F}_p \times \mathbb{F}_d.$$

A square root of -1 in the $\mathbb{F}_p$-factor gives an element whose square is $(-1, 0)$, not $(-1, -1)$. To have a square root of -1 in all of $\mathbb{Z}/n\mathbb{Z}$, one would also need -1 to be a square modulo d, which is not part of the design constraint. What the construction needs is weaker and more targeted: in

$$(\mathbb{Z}/n\mathbb{Z})[i] \simeq \mathbb{F}_p[i] \times \mathbb{F}_d[i],$$

the $\mathbb{F}_p[i]$ factor splits and supplies the nontrivial rotation, while the $\mathbb{F}_d[i]$ factor is kept at the identity. The formal i keeps the real/imaginary bookkeeping uniform across both CRT factors. The implementation chooses an element

$$\omega = \omega_{\mathrm{re}} + i\,\omega_{\mathrm{im}}$$

of order d and norm 1, and defines

$$\cos x = \Re(\omega^{x \bmod d}), \qquad \sin x = \Im(\omega^{x \bmod d}).$$

Complex multiplication immediately gives

$$\begin{aligned} \cos(a+b) &= \cos a \cos b - \sin a \sin b, \\ \sin(a+b) &= \sin a \cos b + \cos a \sin b, \\ \cos^2 x + \sin^2 x &= 1. \end{aligned}$$

This variant gives up perfect cube roots. If $d > 3$ is prime and $p = 4d + 1$ is also prime, then $d \not\equiv 2 \pmod 3$, since that would make $p \equiv 0 \pmod 3$. Thus $d \equiv 1 \pmod 3$, so $3 \mid d - 1$. In the small $d = 3$ case, $3 \mid p - 1$. In either case the unit-group exponent has a factor of 3, so the cube map is not globally invertible by a power map. `cbrt` is therefore represented as a tag in the Pythagorean semantics.

9 Tags and genuinely opaque functions

Not every operation has a useful algebraic law in the target ring. For opaque libdevice calls, miscellaneous transcendentals, and Field-mode exp/log/trig calls, the implementation produces a

deterministic tag. In this document, a tag means the resulting fingerprint, while the operation identifier or symbol hash is the salt used to construct it.

For a k-ary opaque operation symbol f, one should think of the implementation as choosing a deterministic map

$$\tau_f : R^k \longrightarrow R$$

for the current target ring R, extended to the sentinel values by the algebraic NaN rules. It promises congruence:

$$f = g, \quad a_i = b_i \text{ for all } i \quad \Longrightarrow \quad \tau_f(a_1, \ldots, a_k) = \tau_g(b_1, \ldots, b_k).$$

It also uses the operation identity and argument order, so changing the symbol or changing the operands normally changes the fingerprint, up to ordinary finite-ring collisions. What it deliberately does not promise is an algebraic law such as $\log(xy) = \log x + \log y$, unless the semantics has a specific channel implementing that law.

This is not merely a weak fallback. Algebraically, an opaque call is best viewed as adjoining a fresh formal generator $t_{f,\mathbf{a}}$ for the operation and its argument tuple, then continuing the surrounding calculation in the target ring. A fixed-width sanitizer cannot literally adjoin infinitely many independent generators, so the tag is the compact residue fingerprint of that formal-generator idea. The same expression receives the same tag; unrelated opaque expressions are kept distinct except for the unavoidable collision risk of any finite fingerprint.

Triton uses the same opaque-operation idea for operations it does not give a structured algebraic model. For a fixed list of named unary operations, such as `log`, `sqrt`, `erf`, `floor`, and `ceil`, Triton applies a hash-like permutation of the already-scrambled payload, salted by the operation identifier. For arbitrary extern or libdevice calls, it uses a stable hash of the symbol name together with an argument-order-sensitive mix of the operand payloads. These are tags in the same semantic sense as above: deterministic, operation-distinguishing, and deliberately uninterpreted. `hip-fpsan` matches those formulas in `Semantics::Triton`; in the algebraic modes the concrete mixing formula changes to live inside the target ring, but the contract is the same.

No transcendence conjecture is involved in tags. They behave like uninterpreted operation symbols with congruence, not like real transcendental functions. No identity connecting $\tau_f(a_1, \ldots, a_k)$ back to the operands is asserted in the first place.

Once produced, a tag is just another fingerprint. The ambient ring laws still apply to expressions built from it: for example $t + t = 2t$ and $(t + u)v = tv + uv$ are still enforced. The only thing withheld is a mathematical relation between t and the operands of the opaque operation.

10 Infinity, NaN, and order

The finite residue model is extended by one unsigned infinity and one absorbing NaN. In the Field case, the finite residues together with infinity form the projective line

$$\mathbb{P}^1(\mathbb{F}_p) = \mathbb{F}_p \cup \{\infty\},$$

and the full target set of the semantics is

$$\mathbb{P}^1(\mathbb{F}_p) \cup \{\mathrm{NaN}\}.$$

For the composite ring modes, read the same picture as the affine residue ring together with one formal point at infinity, again plus NaN. Correspondingly, the arithmetic rules model the projective line rather than IEEE-754-style signed infinities:

$$x/0 = \infty \quad (x \neq 0), \qquad x/\infty = 0,$$

and indeterminate forms such as $0/0$, $0 \cdot \infty$, and ∞/∞ produce NaN. Since the infinity is unsigned, $\infty + \infty$ also produces NaN in this model. Arithmetic operations on finite residues never overflow; among the basic arithmetic operations on finite values, the only source of infinity is $x/0$ with $x \neq 0$.

No finite ring can carry a total order compatible with its addition and multiplication. Consequently the algebraic semantics do not model IEEE-754 numeric ordering. Comparisons, `min`, and `max` use deterministic fingerprint ordering, as Triton FPSan does.

11 Extension to irrational number schemes

The construction above used one special fact about ordinary binary floating point: finite values are dyadic rationals, hence elements of the localization $\mathbb{Z}[1/2]$. Logarithmic number systems are a long-standing alternative representation [4, 5, 6], and many LNS-like variants make irrational algebraic values exactly representable by allowing fractional binary exponents: half-integral exponents give $\sqrt{2}$, quarter-integral exponents give $2^{1/4}$, and finer grids go further. Such formats may or may not keep a mantissa. Multiplication is then exponent addition, while addition is the hard operation. For FPSan reduction, the only essential change is that the finite values now lie in a number field rather than in $\mathbb{Q}$.

In general, let K be a number field with ring of integers $\mathcal{O}_K$, and suppose the representable finite values lie in a localization $S^{-1}\mathcal{O}_K$, or in an order inside it. For every maximal ideal $\mathfrak{p}$ avoiding S, reduction gives

$$S^{-1}\mathcal{O}_K \longrightarrow \mathcal{O}_K/\mathfrak{p}.$$

The target is a finite field, not necessarily a prime field. Thus "reduce floats mod p" is shorthand for reducing the localized ring of algebraic integers modulo a finite prime. The algebraic number theory used here is standard; see Neukirch [7, Chapter I] for prime ideals, localization, ramification, and the Dedekind–Kummer theorem.

Consider the concrete quarter-exponent example. Let

$$\alpha = 2^{1/4}, \qquad K = \mathbb{Q}(\alpha), \qquad \alpha^4 = 2.$$

A format with scale factors $2^{k/4}$ and a binary mantissa has finite values in the Laurent order

$$\mathbb{Z}[\alpha, \alpha^{-1}].$$

The explicit dyadic localization no longer needs separate notation, because $1/2 = \alpha^{-4}$. This remains true regardless of whether the format has a binary mantissa, and regardless of the mantissa width: mantissa bits introduce only integer coefficients and powers of 2, which are already powers of α. Negative quarter-exponents introduce no new primes in the denominator, because

$$\alpha^{-1} = \alpha^3/2,$$

so inverting α only inverts 2. The polynomial $X^4 - 2$ is Eisenstein at 2, and the discriminant of the basis $1, \alpha, \alpha^2, \alpha^3$ is

$$\operatorname{disc}(X^4 - 2) = 4^4\, 2^3 = 2^{11}.$$

Thus ramification is confined to 2. Every odd prime remains available for reduction, exactly as for dyadic rationals.

For an odd rational prime p, the Dedekind–Kummer theorem reads the decomposition of (p) from the factorization of $X^4 - 2$ over $\mathbb{F}_p$ [7, Chapter I]:

condition on p	$X^4 - 2$ over $\mathbb{F}_p$	residue fields
$p \equiv 1 \pmod 8$, 2 a fourth power	$1+1+1+1$	$\mathbb{F}_p, \mathbb{F}_p, \mathbb{F}_p, \mathbb{F}_p$
$p \equiv 1 \pmod 8$, 2 not a fourth power	$2+2$	$\mathbb{F}_{p^2}, \mathbb{F}_{p^2}$
$p \equiv 7 \pmod 8$	$1+1+2$	$\mathbb{F}_p, \mathbb{F}_p, \mathbb{F}_{p^2}$
$p \equiv 3 \pmod 8$	$2+2$	$\mathbb{F}_{p^2}, \mathbb{F}_{p^2}$
$p \equiv 5 \pmod 8$	4	$\mathbb{F}_{p^4}$

Here $1+1+2$ means two linear factors and one irreducible quadratic factor. The first line is the split case. The last line is the inert case: p remains prime in $\mathbb{Z}[\alpha]$, and the residue field is the non-prime finite field $\mathbb{F}_{p^4}$. The degree-two cases similarly produce residue fields $\mathbb{F}_{p^2}$. Only the completely split case lands entirely in copies of $\mathbb{F}_p$.

For fingerprinting, any of these prime ideals can be a target. If its residue degree is f, the finite target field is $\mathbb{F}_{p^f}$, and the leaf map sends

$$m\,\alpha^k\,2^{-r} \longmapsto \overline{m}\,\overline{\alpha}^{\,k}\,\overline{2}^{-r} \quad \text{in } \mathbb{F}_{p^f}.$$

Since p is odd, $\overline{2}$ is invertible, so the scale factors remain units and the same zero-safety argument applies. This is not implemented here, but the algebraic mechanism is unchanged: ordinary binary floats use the number field $\mathbb{Q}$; quarter-exponent LNS uses $\mathbb{Q}(2^{1/4})$; other algebraic scale systems use other number fields. Maximal ideals not inverted by the format still supply finite fields in which exact algebraic identities can be fingerprinted.

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